

LAB 2 - Linux - NTI

1. List the available shells in your system
cat /etc/shells

2. List all environment variables in your current shell
printenv

3. Display your current shell name

echo \$SHELL

4. List all environment variables for the Bash shell

`bash -c "env"`

5. Edit your shell profile to display date at login and change prompt

<nano ~/.bashrc>

echo "Today is \$(date)"

`export PS1="\u@\h [\\$(date +%H:%M:%S)] \w \$ "`

6. Redirect output of ls command to a file

ls > file_list.txt

7. Use file globbing to list all .txt files

ls *.txt

8. Redirect output of ls command to a file and append it

ls file_list.txt

9. Use pipe to filter ls output for files containing "report"

ls | grep report

10. View first/last 10 lines of a file

```
head filename.txt  
tail filename.txt
```

11. Extract second column from a CSV file

```
cut -d ',' -f 2 data.csv  
### 12. Search for "ERROR" in log.txt
```

```
grep "ERROR" log.txt
```

13. Create shell variable with whoami output

```
current_user=$(whoami)  
echo $current_user
```

14. Convert lowercase to uppercase with tr

```
echo "hello world" | tr '[a-z]' '[A-Z]'
```

15. Search for specific process with ps and grep

```
ps aux | grep process_name
```

16. Create a Bash alias

```
alias ls='ls -l'
```

17. Sort ls output by file size

```
ls -l | sort -k 5 -n
```

18. Count lines containing "success"

```
grep -c "success" filename.txt
```

19. Redirect dmesg output and view first 20 lines

```
dmesg > dmesg_output.txt  
head -n 20 dmesg_output.txt
```

20. Extract first field from CSV file

```
cut -d ',' -f 1 data.csv
```